3.4 Geometrical relationships across languages

Famously, LLMs latent spaces exhibit alluring geometrical properties, notably the parallelogram structures such as `woman:queen::man:king` (Li et al., 2025). Do similar relationships exist at the ensemble level? Tab. 2 suggests that they do. We consider three French novels and their English translations: Gustave Flaubert's *Madame Bovary*, Victor Hugo's *Ninety-Three*, and Émile Zola's *Germinal*. A probe trained to separate a French ensemble pair keeps its accuracy on the corresponding English pair. Similarly, there is a strong similarity between centroid separations in French and in English. The cosine distance between author A and author B in French and in English is between 0.5 and 0.6, which is far smaller than expected for random vectors ($1 \pm 1/\sqrt{2048}$). This observation seems to generalize point-based geometrical structures to distributed clouds of embeddings.

	F/H	F/Z	H/Z
🇫🇷	79%	63%	65%
🇬🇧	82%	65%	63%

Table 2: Accuracy of a reference linear probe trained to distinguish Flaubert (F) from Hugo (H) in French, when applied to other ensemble pairs. The probe achieves about the same accuracy when applied to the corresponding English-translated ensembles. It performs significantly worse (60%) when applied to unrelated pairs involving Zola (Z).

4 Discussion and future directions

Practicalities. We remark that a by-product of training LLMs is that they inherit a fine perception of stylistic and informational patterns, even from short passages. Perhaps literature scholars will build upon this idea to implement more sophisticated methods to address some long-standing mysteries and controversies: Was Shakespeare a single writer? Could Émile Ajar have been identified as Romain Gary before illegitimately snatching a second Prix Goncourt (Tirvengadum, 1996)?

Geometry of style. As an insightful application, we consider the geometry of style partially uncovered in this study and produce a low-dimensional representation. In Fig. 5, we propose a *map of style* where we place various oeuvres based on the relative proximity of their corresponding embeddings. We discuss and interpret this visualization under the lens of literature analysis in Appendix C.

Interpretability and implications. Beyond practicalities, the main objective of this work is to further understand the information content of LLM embeddings. Many studies have revealed that LLMs construct world models in their latent space, allowing encoding of many features in vector representations, often linearly (Park et al., 2024). Some of these features are easily interpretable while others remain obscure (Bricken et al., 2023; Templeton et al., 2024). We have shown here that intangible aspects of an input prompt, namely stylistic features, are also abstractly encoded in deep representations.

5 Conclusion

We have shown that LLM embeddings representing short (10^2 tokens) literary excerpts encode enough information to identify their origin. Increased confusion between books of the same author suggests that embeddings convey a stylistic signature specific to a given writer. This signature appears to lie within a small subspace spanned by the largest principal components. Shuffling words conserved separability, suggesting the main signal might be about lexical content rather than syntax.

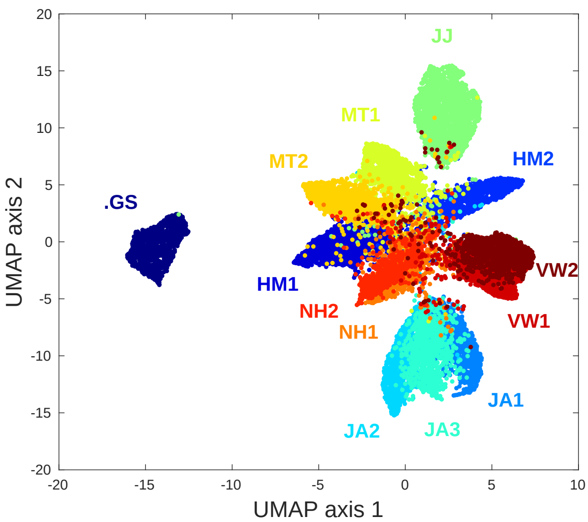

Figure 5: Map of style: low-dimensional visualization of the high-dimensional geometry across books and authors. Text chunks ($N = 128, L = 16$) are UMAP embedded from their 32-dimensional activations extracted at the penultimate layer of the MLP classifier of Fig. 2. We note the substantial overlap between excerpts from the same author, e.g., Austen (**JA1**, **JA2**, **JA3**) or Wolf (**VW1**, **VW2**). More comments in Appendix C.

Acknowledgments

This work was supported by the SciAI Center, and funded by the Office of Naval Research (ONR), under Grant Numbers N00014-23-1-2729 and N00014-23-1-2716.

References

Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, and 3 others. 2025. Smollm2: When smol goes big – data-centric training of a small language model. *Preprint*, arXiv:2502.02737.

Milad Alshomary, Narutatsu Ri, Marianna Apidianaki, Ajay Patel, Smaranda Muresan, and Kathleen McKeown. 2024. Latent space interpretation for stylistic analysis and explainable authorship attribution. *Preprint*, arXiv:2409.07072.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, and 6 others. 2023. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. In *Advances in neural information processing systems*, volume 33, pages 1877–1901.

Tyler A. Chang, Zhuowen Tu, and Benjamin K. Bergen. 2022. The geometry of multilingual language model representations. *Preprint*, arXiv:2205.10964.

Svetlana Gorovaia, Gleb Schmidt, and Ivan P. Yamshchikov. 2024. Sui generis: Large language models for authorship attribution and verification in Latin. In *Proceedings of the 4th International Conference on Natural Language Processing for Digital Humanities*, pages 398–412, Miami, USA. Association for Computational Linguistics.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Wes Gurnee and Max Tegmark. 2024. Language models represent space and time. *Preprint*, arXiv:2310.02207.

Rebecca M. M. Hicke and David Mimno. 2023. T5 meets tybalt: Author attribution in early modern english drama using large language models. *Preprint*, arXiv:2310.18454.

Baixiang Huang, Canyu Chen, and Kai Shu. 2025. Authorship attribution in the era of llms: Problems, methodologies, and challenges. *Preprint*, arXiv:2408.08946.

Artem Kirsanov, Chi-Ning Chou, Kyunghyun Cho, and SueYeon Chung. 2025. The geometry of prompting: Unveiling distinct mechanisms of task adaptation in language models. *Preprint*, arXiv:2502.08009.

Yuxiao Li, Eric J. Michaud, David D. Baek, Joshua Engels, Xiaoqing Sun, and Max Tegmark. 2025. The geometry of concepts: Sparse autoencoder feature structure. *Entropy*, 27(4).

Qing Lyu, Marianna Apidianaki, and Chris Callison-burch. 2023. Representation of lexical stylistic features in language models' embedding space. In *Proceedings of the 12th Joint Conference on Lexical and Computational Semantics (*SEM 2023)*, pages 370–387, Toronto, Canada. Association for Computational Linguistics.

Samuel Marks and Max Tegmark. 2024. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *Preprint*, arXiv:2310.06824.

MetaAI. 2024. Llama 3.2 model card. `https://github.com/meta-llama/llama-models/blob/main/models/llama3_2/MODEL_CARD.md`. Accessed: 2024-09-25.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013a. Efficient estimation of word representations in vector space. *Preprint*, arXiv:1301.3781.

Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013b. Distributed representations of words and phrases and their compositionality. In *Advances in neural information processing systems*, volume 26.

Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. 2025. The geometry of categorical and hierarchical concepts in large language models. *Preprint*, arXiv:2406.01506.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2024. The linear representation hypothesis and the geometry of large language models. *Preprint*, arXiv:2311.03658.

Ajay Patel, Delip Rao, Ansh Kothary, Kathleen McKeown, and Chris Callison-Burch. 2023. Learning interpretable style embeddings via prompting LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 15270–15290, Singapore. Association for Computational Linguistics.

Raphaël Sarfati, Toni J. B. Liu, Nicolas Boullé, and Christopher J. Earls. 2025. Lines of thought in large language models. *Preprint*, arXiv:2410.01545.

William Shakespeare. ca. 1599. *The Tragedy of Romeo and Juliet*. Folger Shakespeare Library. Accessed: 2025-04-07.

Adi Simhi and Shaul Markovitch. 2023. Interpreting embedding spaces by conceptualization. In *The 2023 Conference on Empirical Methods in Natural Language Processing*.

Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, Thomas Mesnard, Geoffrey Cideron, Jean bastien Grill, Sabela Ramos, Edouard Yvinec, Michelle Casbon, Etienne Pot, Ivo Penchev, and 197 others. 2025. Gemma 3 technical report. *Preprint*, arXiv:2503.19786.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, and 3 others. 2024. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*.

V. Tirvengadum. 1996. Linguistic fingerprints and literary fraud. *Digital Studies/le Champ Numérique*, 6.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, and 1 others. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Lucrezia Valeriani, Diego Doimo, Francesca Cuturello, Alessandro Laio, Alessio Ansuini, and Alberto Cazzaniga. 2023. The geometry of hidden representations of large transformer models. *Preprint*, arXiv:2302.00294.

Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, and Matteo Biagetti. 2025. The geometry of tokens in internal representations of large language models. *Preprint*, arXiv:2501.10573.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

A Additional background

A.1 LLM internal geometry

LLMs have been found to encode high-level attributes in surprising geometric patterns within their embedding spaces. Recent work supports a linear representation hypothesis, wherein certain abstract concepts correspond to linear directions or subspaces in the latent representation space (Park et al., 2024). For instance, models appear to linearly represent attributes like factual truthfulness, enabling simple probes or even direct manipulation of activations along those concept directions (Marks and Tegmark, 2024). Similarly, categorical semantic relationships can emerge as geometric structures: models represent categories as vertices of a simplex and encode hierarchical relations via approximately orthogonal components (Park et al., 2025). The representational geometry induced by prompts has also been analyzed; different prompting or in-context learning strategies imprint distinct geometric signatures on a model's internal states, highlighting how task framing can alter the organization of latent features (Kirsanov et al., 2025). Moreover, in multilingual settings, LLM embedding spaces can separate language-specific style from language-neutral content along perpendicular axes, suggesting a degree of factorization between surface form and underlying meaning (Chang et al., 2022).

A.2 Authorship attribution

An important high-level attribute of interest is literary style. Authorship attribution and stylistic analysis have served as tests for whether models capture subtle distributional differences beyond topic or semantics. Traditional stylometry relied on carefully engineered linguistic features (e.g. function word frequencies, character n-grams, syntactic patterns), but modern transformer-based LMs can learn such distinctions directly from raw text (Hicke and Mimno, 2023). Recent studies demonstrate that large pretrained models achieve strong performance on author identification. For example, Hicke and Mimno (2023) showed that a fine-tuned T5 model can attribute Early Modern English plays to their likely authors, indicating that LLM representations encode distinctive stylistic signatures. Likewise, GPT-based methods have been applied to Latin prose to verify authorship, with results rivaling traditional stylometric classifiers (Gorovaia et al., 2024). Notably, these analy-